MSG No. 227.130 $Fd'\bar{3}'m$ [Type III, cubic]

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det	TR
1	$\{1\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
2	$\{2_{100}\|0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	1	1
3	$\{2_{010}\|\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{4} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	1	1
4	$\{2_{001}\|\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
5	$\{3^+_{111}\|0\}$	$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$	1	1
6	$\{3^-_{111}\|0\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$	1	1
7	$\{3^-_{1-1-1}\|\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	1	1
8	$\{3^+_{1-1-1}\|\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & 1 & 0 & 0 \end{bmatrix}$	1	1
9	$\{3^-_{-11-1}\|\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{1}{4} \\ 1 & 0 & 0 & 0 \end{bmatrix}$	1	1
10	$\{3^+_{-11-1}\|0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{1}{4} \end{bmatrix}$	1	1
11	$\{3^-_{-1-11}\|0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	1	1
12	$\{3^+_{-1-11}\|\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{4} \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{4} \end{bmatrix}$	1	1
13	$\{-4^+_{100}\|\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & \frac{1}{4} \end{bmatrix}$	−1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
14	$\{-4^{-}_{100}\vert\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{1}{4} \\ 0 & 1 & 0 & 0 \end{bmatrix}$	-1	1
15	$\{-4^{+}_{010}\vert\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{1}{4} \\ 1 & 0 & 0 & 0 \end{bmatrix}$	-1	1
16	$\{-4^{-}_{010}\vert 0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	-1	1
17	$\{-4^{+}_{001}\vert 0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	-1	1
18	$\{-4^{-}_{001}\vert\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	-1	1
19	$\{\mathrm{m}_{110}\vert\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	-1	1
20	$\{\mathrm{m}_{1-10}\vert 0\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	-1	1
21	$\{\mathrm{m}_{011}\vert 0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{1}{4} \end{bmatrix}$	-1	1
22	$\{\mathrm{m}_{01-1}\vert 0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$	-1	1
23	$\{\mathrm{m}_{101}\vert\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{4} \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	-1	1
24	$\{\mathrm{m}_{-101}\vert 0\}$	$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$	-1	1
25	$\{4^{+}_{100}{}'\vert\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & \frac{1}{4} \end{bmatrix}$	1	-1
26	$\{4^{-}_{100}{}'\vert\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{4} \\ 0 & -1 & 0 & 0 \end{bmatrix}$	1	-1
27	$\{4^{+}_{010}{}'\vert\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{4} \\ -1 & 0 & 0 & 0 \end{bmatrix}$	1	-1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
28	$\{4^-_{010}{}'\|0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	1	−1
29	$\{4^+_{001}{}'\|0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	1	−1
30	$\{4^-_{001}{}'\|\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{4} \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	1	−1
31	$\{2_{110}{}'\|\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	−1
32	$\{2_{1-10}{}'\|0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	1	−1
33	$\{2_{011}{}'\|0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{4} \end{bmatrix}$	1	−1
34	$\{2_{01-1}{}'\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$	1	−1
35	$\{2_{101}{}'\|\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{4} \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	1	−1
36	$\{2_{-101}{}'\|0\}$	$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix}$	1	−1
37	$\{-1'\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	−1
38	$\{\mathrm{m}_{100}{}'\|0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	−1	−1
39	$\{\mathrm{m}_{010}{}'\|\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{4} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	−1	−1
40	$\{\mathrm{m}_{001}{}'\|\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	−1
41	$\{-3^+_{111}{}'\|0\}$	$\begin{bmatrix} 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$	−1	−1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
42	$\{-3^{-}_{111}{}'\|0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix}$	-1	-1
43	$\{-3^{-}_{1-1-1}{}'\|\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	-1	-1
44	$\{-3^{+}_{1-1-1}{}'\|\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & -1 & 0 & 0 \end{bmatrix}$	-1	-1
45	$\{-3^{-}_{-11-1}{}'\|\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{4} \\ -1 & 0 & 0 & 0 \end{bmatrix}$	-1	-1
46	$\{-3^{+}_{-11-1}{}'\|0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{4} \end{bmatrix}$	-1	-1
47	$\{-3^{-}_{-1-11}{}'\|0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	-1	-1
48	$\{-3^{+}_{-1-11}{}'\|\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{4} \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{4} \end{bmatrix}$	-1	-1
49	$\{1\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1
50	$\{2_{100}\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	1	1
51	$\{2_{010}\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	1	1
52	$\{2_{001}\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1
53	$\{3^{+}_{111}\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \end{bmatrix}$	1	1
54	$\{3^{-}_{111}\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	1	1
55	$\{3^{-}_{1-1-1}\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
56	$\{3^{+}_{1-1-1}\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{3}{4} \\ 0 & 1 & 0 & \frac{1}{2} \end{bmatrix}$	1	1
57	$\{3^{-}_{-11-1}\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	1	1
58	$\{3^{+}_{-11-1}\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & \frac{3}{4} \\ 0 & -1 & 0 & \frac{3}{4} \end{bmatrix}$	1	1
59	$\{3^{-}_{-1-11}\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	1	1
60	$\{3^{+}_{-1-11}\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{3}{4} \end{bmatrix}$	1	1
61	$\{-4^{+}_{100}\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{3}{4} \end{bmatrix}$	−1	1
62	$\{-4^{-}_{100}\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{3}{4} \\ 0 & 1 & 0 & \frac{1}{2} \end{bmatrix}$	−1	1
63	$\{-4^{+}_{010}\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	−1	1
64	$\{-4^{-}_{010}\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	−1	1
65	$\{-4^{+}_{001}\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	−1	1
66	$\{-4^{-}_{001}\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	−1	1
67	$\{\mathrm{m}_{110}\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	−1	1
68	$\{\mathrm{m}_{1-10}\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	−1	1
69	$\{\mathrm{m}_{011}\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & \frac{3}{4} \\ 0 & -1 & 0 & \frac{3}{4} \end{bmatrix}$	−1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
70	$\{\mathrm{m}_{01-1}\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}1&0&0&0\\0&0&1&\frac{1}{2}\\0&1&0&\frac{1}{2}\end{bmatrix}$	-1	1
71	$\{\mathrm{m}_{101}\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix}0&0&-1&\frac{1}{4}\\0&1&0&\frac{1}{2}\\-1&0&0&\frac{3}{4}\end{bmatrix}$	-1	1
72	$\{\mathrm{m}_{-101}\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}0&0&1&0\\0&1&0&\frac{1}{2}\\1&0&0&\frac{1}{2}\end{bmatrix}$	-1	1
73	$\{4^{+}_{100}{}'\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix}1&0&0&\frac{1}{4}\\0&0&-1&\frac{1}{2}\\0&1&0&\frac{3}{4}\end{bmatrix}$	1	-1
74	$\{4^{-}_{100}{}'\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix}1&0&0&\frac{1}{4}\\0&0&1&\frac{3}{4}\\0&-1&0&\frac{1}{2}\end{bmatrix}$	1	-1
75	$\{4^{+}_{010}{}'\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix}0&0&1&\frac{1}{4}\\0&1&0&\frac{3}{4}\\-1&0&0&\frac{1}{2}\end{bmatrix}$	1	-1
76	$\{4^{-}_{010}{}'\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix}0&0&-1&0\\0&1&0&\frac{3}{4}\\1&0&0&\frac{3}{4}\end{bmatrix}$	1	-1
77	$\{4^{+}_{001}{}'\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix}0&-1&0&0\\1&0&0&\frac{3}{4}\\0&0&1&\frac{3}{4}\end{bmatrix}$	1	-1
78	$\{4^{-}_{001}{}'\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix}0&1&0&\frac{1}{4}\\-1&0&0&\frac{1}{2}\\0&0&1&\frac{3}{4}\end{bmatrix}$	1	-1
79	$\{2_{110}{}'\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix}0&1&0&\frac{1}{4}\\1&0&0&\frac{3}{4}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	1	-1
80	$\{2_{1-10}{}'\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}0&-1&0&0\\-1&0&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	1	-1
81	$\{2_{011}{}'\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix}-1&0&0&0\\0&0&1&\frac{3}{4}\\0&1&0&\frac{3}{4}\end{bmatrix}$	1	-1
82	$\{2_{01-1}{}'\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&0\\0&0&-1&\frac{1}{2}\\0&-1&0&\frac{1}{2}\end{bmatrix}$	1	-1
83	$\{2_{101}{}'\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix}0&0&1&\frac{1}{4}\\0&-1&0&\frac{1}{2}\\1&0&0&\frac{3}{4}\end{bmatrix}$	1	-1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
84	$\{2_{-101}'\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	1	−1
85	$\{-1'\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	−1	−1
86	$\{\mathrm{m}_{100}'\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}$	−1	−1
87	$\{\mathrm{m}_{010}'\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}$	−1	−1
88	$\{\mathrm{m}_{001}'\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	−1	−1
89	$\{-3^{+}_{111}'\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \end{bmatrix}$	−1	−1
90	$\{-3^{-}_{111}'\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	−1	−1
91	$\{-3^{-}_{1-1-1}'\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	−1	−1
92	$\{-3^{+}_{1-1-1}'\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{3}{4} \\ 0 & -1 & 0 & \frac{1}{2} \end{bmatrix}$	−1	−1
93	$\{-3^{-}_{-11-1}'\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	−1	−1
94	$\{-3^{+}_{-11-1}'\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & \frac{3}{4} \\ 0 & 1 & 0 & \frac{3}{4} \end{bmatrix}$	−1	−1
95	$\{-3^{-}_{-1-11}'\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	−1	−1
96	$\{-3^{+}_{-1-11}'\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{3}{4} \end{bmatrix}$	−1	−1
97	$\{1\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
98	$\{2_{100}\|\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	1	1
99	$\{2_{010}\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{3}{4} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	1	1
100	$\{2_{001}\|\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{3}{4} \\ 0 & -1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1
101	$\{3^+_{111}\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \end{bmatrix}$	1	1
102	$\{3^-_{111}\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	1	1
103	$\{3^-_{1-1-1}\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	1	1
104	$\{3^+_{1-1-1}\|\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{2} \end{bmatrix}$	1	1
105	$\{3^-_{-11-1}\|\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	1	1
106	$\{3^+_{-11-1}\|\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{3}{4} \end{bmatrix}$	1	1
107	$\{3^-_{-1-11}\|\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	1	1
108	$\{3^+_{-1-11}\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{3}{4} \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{3}{4} \end{bmatrix}$	1	1
109	$\{-4^+_{100}\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & \frac{3}{4} \end{bmatrix}$	−1	1
110	$\{-4^-_{100}\|\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{2} \end{bmatrix}$	−1	1
111	$\{-4^+_{010}\|\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{3}{4} \\ 0 & -1 & 0 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	−1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
112	$\{-4^{-}_{010}\|\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	-1	1
113	$\{-4^{+}_{001}\|\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	-1	1
114	$\{-4^{-}_{001}\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{3}{4} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	-1	1
115	$\{\mathrm{m}_{110}\|\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	-1	1
116	$\{\mathrm{m}_{1-10}\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	-1	1
117	$\{\mathrm{m}_{011}\|\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{3}{4} \end{bmatrix}$	-1	1
118	$\{\mathrm{m}_{01-1}\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \end{bmatrix}$	-1	1
119	$\{\mathrm{m}_{101}\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{3}{4} \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	-1	1
120	$\{\mathrm{m}_{-101}\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	-1	1
121	$\{4^{+}_{100}{}'\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & \frac{3}{4} \end{bmatrix}$	1	-1
122	$\{4^{-}_{100}{}'\|\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{1}{2} \end{bmatrix}$	1	-1
123	$\{4^{+}_{010}{}'\|\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{3}{4} \\ 0 & 1 & 0 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	1	-1
124	$\{4^{-}_{010}{}'\|\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	1	-1
125	$\{4^{+}_{001}{}'\|\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}$	1	-1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
126	$\{4^-_{001}{}'\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{3}{4} \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}$	1	−1
127	$\{2_{110}{}'\|\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	1	−1
128	$\{2_{1-10}{}'\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	1	−1
129	$\{2_{011}{}'\|\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{3}{4} \end{bmatrix}$	1	−1
130	$\{2_{01-1}{}'\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \end{bmatrix}$	1	−1
131	$\{2_{101}{}'\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{3}{4} \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	1	−1
132	$\{2_{-101}{}'\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{2} \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	1	−1
133	$\{-1'\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	−1	−1
134	$\{\mathrm{m}_{100}{}'\|\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}$	−1	−1
135	$\{\mathrm{m}_{010}{}'\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{3}{4} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}$	−1	−1
136	$\{\mathrm{m}_{001}{}'\|\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{3}{4} \\ 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	−1	−1
137	$\{-3^+_{111}{}'\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{2} \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \end{bmatrix}$	−1	−1
138	$\{-3^-_{111}{}'\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	−1	−1
139	$\{-3^-_{1-1-1}{}'\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	−1	−1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
140	$\{-3^{+}_{1-1-1}{}'\vert\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{1}{2} \end{bmatrix}$	-1	-1
141	$\{-3^{-}_{-11-1}{}'\vert\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{4} \\ -1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$	-1	-1
142	$\{-3^{+}_{-11-1}{}'\vert\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{3}{4} \end{bmatrix}$	-1	-1
143	$\{-3^{-}_{-1-11}{}'\vert\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{4} \\ 1 & 0 & 0 & \frac{3}{4} \end{bmatrix}$	-1	-1
144	$\{-3^{+}_{-1-11}{}'\vert\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{3}{4} \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{3}{4} \end{bmatrix}$	-1	-1
145	$\{1\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
146	$\{2_{100}\vert\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	1	1
147	$\{2_{010}\vert\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{3}{4} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	1	1
148	$\{2_{001}\vert\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{3}{4} \\ 0 & -1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
149	$\{3^{+}_{111}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 \end{bmatrix}$	1	1
150	$\{3^{-}_{111}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \\ 1 & 0 & 0 & 0 \end{bmatrix}$	1	1
151	$\{3^{-}_{1-1-1}\vert\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	1	1
152	$\{3^{+}_{1-1-1}\vert\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{3}{4} \\ 0 & 1 & 0 & 0 \end{bmatrix}$	1	1
153	$\{3^{-}_{-11-1}\vert\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{3}{4} \\ 1 & 0 & 0 & 0 \end{bmatrix}$	1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
154	$\{3^{+}_{-11-1}\|\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{3}{4} \\ 0 & -1 & 0 & \frac{1}{4} \end{bmatrix}$	1	1
155	$\{3^{-}_{-1-11}\|\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	1	1
156	$\{3^{+}_{-1-11}\|\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{4} \end{bmatrix}$	1	1
157	$\{-4^{+}_{100}\|\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{4} \end{bmatrix}$	−1	1
158	$\{-4^{-}_{100}\|\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{3}{4} \\ 0 & 1 & 0 & 0 \end{bmatrix}$	−1	1
159	$\{-4^{+}_{010}\|\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{3}{4} \\ 0 & -1 & 0 & \frac{3}{4} \\ 1 & 0 & 0 & 0 \end{bmatrix}$	−1	1
160	$\{-4^{-}_{010}\|\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	−1	1
161	$\{-4^{+}_{001}\|\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	−1	1
162	$\{-4^{-}_{001}\|\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	−1	1
163	$\{\mathrm{m}_{110}\|\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	−1	1
164	$\{\mathrm{m}_{1-10}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	−1	1
165	$\{\mathrm{m}_{011}\|\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{3}{4} \\ 0 & -1 & 0 & \frac{1}{4} \end{bmatrix}$	−1	1
166	$\{\mathrm{m}_{01-1}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 1 & 0 & 0 \end{bmatrix}$	−1	1
167	$\{\mathrm{m}_{101}\|\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{3}{4} \\ 0 & 1 & 0 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	−1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
168	$\{\mathrm{m}_{-101}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}0&0&1&\frac{1}{2}\\0&1&0&\frac{1}{2}\\1&0&0&0\end{bmatrix}$	-1	1
169	$\{4^{+}_{100}{}'\vert\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix}1&0&0&\frac{3}{4}\\0&0&-1&\frac{1}{2}\\0&1&0&\frac{1}{4}\end{bmatrix}$	1	-1
170	$\{4^{-}_{100}{}'\vert\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix}1&0&0&\frac{3}{4}\\0&0&1&\frac{3}{4}\\0&-1&0&0\end{bmatrix}$	1	-1
171	$\{4^{+}_{010}{}'\vert\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix}0&0&1&\frac{3}{4}\\0&1&0&\frac{3}{4}\\-1&0&0&0\end{bmatrix}$	1	-1
172	$\{4^{-}_{010}{}'\vert\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix}0&0&-1&\frac{1}{2}\\0&1&0&\frac{3}{4}\\1&0&0&\frac{1}{4}\end{bmatrix}$	1	-1
173	$\{4^{+}_{001}{}'\vert\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix}0&-1&0&\frac{1}{2}\\1&0&0&\frac{3}{4}\\0&0&1&\frac{1}{4}\end{bmatrix}$	1	-1
174	$\{4^{-}_{001}{}'\vert\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix}0&1&0&\frac{3}{4}\\-1&0&0&\frac{1}{2}\\0&0&1&\frac{1}{4}\end{bmatrix}$	1	-1
175	$\{2_{110}{}'\vert\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix}0&1&0&\frac{3}{4}\\1&0&0&\frac{3}{4}\\0&0&-1&0\end{bmatrix}$	1	-1
176	$\{2_{1-10}{}'\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}0&-1&0&\frac{1}{2}\\-1&0&0&\frac{1}{2}\\0&0&-1&0\end{bmatrix}$	1	-1
177	$\{2_{011}{}'\vert\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&0&1&\frac{3}{4}\\0&1&0&\frac{1}{4}\end{bmatrix}$	1	-1
178	$\{2_{01-1}{}'\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\\0&-1&0&0\end{bmatrix}$	1	-1
179	$\{2_{101}{}'\vert\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix}0&0&1&\frac{3}{4}\\0&-1&0&\frac{1}{2}\\1&0&0&\frac{1}{4}\end{bmatrix}$	1	-1
180	$\{2_{-101}{}'\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}0&0&-1&\frac{1}{2}\\0&-1&0&\frac{1}{2}\\-1&0&0&0\end{bmatrix}$	1	-1
181	$\{-1'\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&-1&0&\frac{1}{2}\\0&0&-1&0\end{bmatrix}$	-1	-1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
182	$\{\mathrm{m}_{100}{}'\vert\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	-1	-1
183	$\{\mathrm{m}_{010}{}'\vert\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{3}{4} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	-1	-1
184	$\{\mathrm{m}_{001}{}'\vert\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{3}{4} \\ 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	-1	-1
185	$\{-3^{+}_{111}{}'\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & 0 \end{bmatrix}$	-1	-1
186	$\{-3^{-}_{111}{}'\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{2} \\ -1 & 0 & 0 & 0 \end{bmatrix}$	-1	-1
187	$\{-3^{-}_{1-1-1}{}'\vert\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	-1	-1
188	$\{-3^{+}_{1-1-1}{}'\vert\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{3}{4} \\ 0 & -1 & 0 & 0 \end{bmatrix}$	-1	-1
189	$\{-3^{-}_{-11-1}{}'\vert\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{3}{4} \\ -1 & 0 & 0 & 0 \end{bmatrix}$	-1	-1
190	$\{-3^{+}_{-11-1}{}'\vert\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & -1 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{3}{4} \\ 0 & 1 & 0 & \frac{1}{4} \end{bmatrix}$	-1	-1
191	$\{-3^{-}_{-1-11}{}'\vert\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{3}{4} \\ 1 & 0 & 0 & \frac{1}{4} \end{bmatrix}$	-1	-1
192	$\{-3^{+}_{-1-11}{}'\vert\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix} 0 & 0 & 1 & \frac{3}{4} \\ -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{4} \end{bmatrix}$	-1	-1